

11.5 We are given:

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$$

$$\mathbf{y} = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}$$

$$\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$$

$\mathbf{L}$ is the Cholesky factor here, and since all the diagonal values are strictly positive real numbers, it's determinant is non-zero, which proves that $\mathbf{L}$ is invertible.

Now we use the change of variable technique specified in (6.143) in the book Mathematics for Machine Learning:

$$p(\mathbf{y}) = p(U^{-1}(\mathbf{y})) \cdot \left| \frac{d}{d\mathbf{y}} U^{-1}(\mathbf{y}) \right|$$

where $U(\mathbf{z}) = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}$ is an invertible transformation, and $U^{-1}(\mathbf{y}) = \mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu})$.

$$p(\mathbf{y}) = p(\mathbf{z}) \cdot \left| \frac{d\mathbf{z}}{d\mathbf{y}} \right|$$

$$\frac{d\mathbf{y}}{d\mathbf{z}} = \frac{d(\boldsymbol{\mu} + \mathbf{L}\mathbf{z})}{d\mathbf{z}} = \mathbf{L} \quad \text{and} \quad |\mathbf{L}| = |\boldsymbol{\Sigma}|^{1/2}$$

$$\implies \left| \frac{d\mathbf{z}}{d\mathbf{y}} \right| = |\boldsymbol{\Sigma}|^{-1/2}$$

$$\implies p(\mathbf{y}) = |\boldsymbol{\Sigma}|^{-1/2} \cdot p(\mathbf{z})$$

$$= |\boldsymbol{\Sigma}|^{-1/2} \cdot \frac{1}{\sqrt{(2\pi)^{D/2}}} \exp \left\{ -\frac{\mathbf{z}^T \mathbf{z}}{2} \right\}$$

$$= \frac{1}{\sqrt{(2\pi)^{D/2} |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{\mathbf{z}^T \mathbf{z}}{2} \right\}$$

$$= \frac{1}{\sqrt{(2\pi)^{D/2} |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{(\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu}))^T (\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu}))}{2} \right\}$$

Simplifying the numerator term in the exponent:

$$\begin{aligned}
& (\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu}))^T (\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu})) \\
&= (\mathbf{L}^{-1}\mathbf{y} - \mathbf{L}^{-1}\boldsymbol{\mu})^T (\mathbf{L}^{-1}\mathbf{y} - \mathbf{L}^{-1}\boldsymbol{\mu}) \\
&= (\mathbf{y}^T \mathbf{L}^{-T} - \boldsymbol{\mu}^T \mathbf{L}^{-T})(\mathbf{L}^{-1}\mathbf{y} - \mathbf{L}^{-1}\boldsymbol{\mu}) \\
&= \mathbf{y}^T \mathbf{L}^{-T} \mathbf{L}^{-1} \mathbf{y} - \mathbf{y}^T \mathbf{L}^{-T} \mathbf{L}^{-1} \boldsymbol{\mu} - \boldsymbol{\mu}^T \mathbf{L}^{-T} \mathbf{L}^{-1} \mathbf{y} + \boldsymbol{\mu}^T \mathbf{L}^{-T} \mathbf{L}^{-1} \boldsymbol{\mu}
\end{aligned}$$

Simplifying $\mathbf{L}^{-T} \mathbf{L}^{-1}$,

$$\mathbf{L}^{-T} \mathbf{L}^{-1} = (\mathbf{L} \mathbf{L}^T)^{-1} = \boldsymbol{\Sigma}^{-1}$$

Substituting this result back, we get:

$$\begin{aligned}
&= \mathbf{y}^T \boldsymbol{\Sigma}^{-1} \mathbf{y} - \mathbf{y}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} - \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \mathbf{y} + \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} \\
&= (\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})
\end{aligned}$$

Putting this back into the numerator of the exponent:

$$\begin{aligned}
&= \frac{1}{\sqrt{(2\pi)^{D/2} |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{(\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})}{2} \right\} \\
&= \frac{1}{\sqrt{(2\pi)^{D/2} |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{(\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})}{2} \right\} \\
&= \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})
\end{aligned}$$